CHURNQUEST SOLUTION GUIDE

1. Executive Summary

Customer churn is a significant challenge in the telecommunications industry, affecting customer retention, recurring revenue, and long-term business performance.

This project analyzes a telecommunications customer dataset containing 4,250 customer records to identify patterns associated with customer churn.

The analysis was performed using Microsoft Excel, Power Query, PivotTables, and data visualizations.

The results reveal an **overall customer churn rate of 14.07%**, representing 598 churned customers and 3,652 retained customers.

The most significant findings include:

- Customers making four or more customer service calls have a churn rate of 50.75%, compared with 10.93% for customers making fewer calls.
- International plan subscribers have a churn rate of 42.17%, compared with 11.18% for non-subscribers.
- Customers without voicemail subscriptions have a churn rate of 16.44%, compared with 7.37% for voicemail subscribers.
- Churned customers average 208.99 daytime calling minutes, compared with 175.56 minutes for retained customers.
- New Jersey has the highest observed state-level churn rate at 27.08%.
- The highest-risk observed customer segment consists of customers with four or more service calls, an international plan, and no voicemail plan, with a churn rate of 79.17%.

These findings indicate that customer service interactions, international plan subscriptions, and calling behavior are important characteristics to consider when developing customer retention strategies.

The project concludes with recommendations for proactive customer support, international plan evaluation, targeted customer engagement, and further investigation of high-risk customer segments.

2. Dataset Overview

The dataset contains information about 4,250 telecommunications customers.

Each record represents an individual customer and includes information related to account characteristics, service subscriptions, calling behavior, charges, customer service interactions, and churn status.

2.1 Dataset Summary

Attribute	Description
Total Records	4,250
Original Columns	20
Data Format	CSV
Analysis Platform	Microsoft Excel
Data Cleaning Tool	Power Query
Target Variable	Churn
Churn Categories	Yes / No
Geographic Coverage	US States

2.2 Key Variables

The analysis uses the following variables:

- Account Length
- State
- International Plan
- Voice Mail Plan
- Customer Service Calls
- Total Day Minutes
- Total Evening Minutes
- Total Night Minutes
- Total International Minutes
- Total Day Charge
- Total Evening Charge
- Total Night Charge
- Total International Charge
- Churn

Additional variables were created during data preparation to support segmentation and churn calculations.

---

3. Data Cleaning and Preparation

The dataset was imported into Microsoft Excel and transformed using Power Query.

3.1 Data Quality Assessment

The initial dataset was reviewed for missing values, duplicate records, incorrect data types, and inconsistent categorical values.

The assessment identified:

- Total Records: 4,250
- Missing Values: None identified
- Duplicate Records: None identified

Appropriate data types were assigned to numerical and categorical columns.

3.2 Data Transformation

Three additional columns were created.

A. Churn_Flag

A binary indicator was created to simplify churn calculations.

- Churn = Yes → 1
- Churn = No → 0

This field was used to calculate churned customer counts and churn rates.

B. Tenure_Group

Customers were categorized based on account length:

Account Length	**Tenure Group**
Less than 24 months	0–23 Months
24–59 months	24–59 Months
60–95 months	60–95 Months
96–131 months	96–131 Months
132 months or more	132+ Months

C. Service_Call_Group

Customers were categorized according to customer service interactions:

Customer Service Calls	Group
0–3	0–3 Calls
4 or more	4+ Calls

These transformed variables were used to conduct customer segmentation analysis.

4. Analytical Methodology

The project follows a descriptive and exploratory data analysis approach.

Microsoft Excel PivotTables were used to calculate customer counts, churn rates, average calling minutes, and average charges.

4.1 Key Calculations

Total Customers: Count of customer records.

Churned Customers: Sum of Churn_Flag.

Retained Customers: Total Customers − Churned Customers.

Churn Rate: Churn Rate (%) = (Churned Customers / Total Customers) × 100

Average Calling Minutes: Total calling minutes within a customer group divided by the number of customers in that group.

Average Charges: Total charges within a customer group divided by the number of customers in that group.

4.2 PivotTable Calculations

The Churn_Flag variable was summarized using three operations:

- Count: Total customers
- Sum: Churned customers
- Average: Churn rate

Because Churn_Flag contains only 0 and 1, its average represents the proportion of customers who churned.

5. Detailed Analysis and Findings

ANALYSIS 1: Overall Customer Churn

Business Question

What is the overall customer churn rate?

Methodology

A PivotTable was created using the churn variable to count retained and churned customers.

Customer Status	Customers	Percentage
Retained	3,652	85.93%
Churned	598	14.07%
Total	4,250	100.00%

Key Findings

- The overall churn rate is **14.07%**.
- This means approximately 14 out of every 100 customers in the dataset discontinued their service.
- The majority of customers, representing 85.93%, remained with the company.

Business Interpretation

- Although most customers were retained, the churned customer population represents an important opportunity for improving customer retention.
- Identifying characteristics associated with churn can help the company prioritize further investigation and retention initiatives.

ANALYSIS 2: Customer Tenure vs. Churn

Business Question

How does customer tenure relate to churn?

Methodology

Average account length was calculated separately for churned and retained customers.

Customers were also divided into five tenure groups to compare churn rates.

Results: Average Tenure

Customer Status	Average Tenure
Retained	99.92 months
Churned	102.14 months

Results: Tenure Groups

Tenure Group	Customers	Churn Rate
0–23 Months	104	9.62%
24–59 Months	556	11.69%
60–95 Months	1,301	14.37%
96–131 Months	1,380	15.36%
132+ Months	909	13.64%

Key Findings

- Churned customers have an average account length approximately 2.22 months greater than retained customers.
- The highest observed churn rate occurs in the 96–131 months tenure group at 15.36%.
- The lowest observed churn rate occurs in the 0–23 months group at 9.62%.

Business Interpretation

- The results do not show a consistent decline in churn as customer tenure increases.
- This suggests that longer account tenure does not necessarily correspond to lower churn in the dataset.
- However, the newest tenure group contains only 104 customers, so its churn rate should be interpreted cautiously.

ANALYSIS 3: International Plan vs. Churn

Business Question

Does international plan subscription relate to customer churn?

Methodology

Customers were grouped according to international plan subscription status.

Customer counts, churned counts, and churn rates were calculated.

Results

International Plan	Customers	Churned	Churn Rate
No	3,854	431	11.18%
Yes	396	167	42.17%
Total	4,250	598	14.07%

Key Findings

- Customers with international plans have a churn rate of 42.17%.
- Customers without international plans have a churn rate of 11.18%.
- The difference is approximately 30.99 percentage points.
- International plan subscribers exhibit approximately 3.77 times the churn rate of non-subscribers.

Business Interpretation

- International plan subscription is strongly associated with higher customer churn.
- Possible areas for further investigation include international calling charges, plan benefits, service quality, and customer expectations.
- The analysis does not establish that subscribing to an international plan causes churn.

ANALYSIS 4: Voicemail Plan vs. Churn

Business Question

How does voicemail subscription relate to customer churn?

Methodology

Customers were grouped according to voicemail plan subscription status.

Results

Voicemail Plan	Customers	Churned	Churn Rate
No	3,138	516	16.44%
Yes	1,112	82	7.37%
Total	4,250	598	14.07%

Key Findings

- Customers without voicemail plans have a churn rate of 16.44%.
- Customers with voicemail plans have a churn rate of 7.37%.
- The difference is approximately 9.07 percentage points.

Business Interpretation

- Voicemail subscription is associated with lower observed churn.
- This may reflect differences in customer engagement, service needs, or other characteristics.
- The results do not establish that voicemail subscriptions directly prevent customer churn.

ANALYSIS 5: Customer Service Calls vs. Churn

Business Question

Are customers making four or more customer service calls more likely to churn?

Methodology

Customers were divided into two groups based on the number of customer service calls.

Results

Service Calls	Customers	Churned	Churn Rate
0–3 Calls	3,915	428	10.93%
4+ Calls	335	170	50.75%
Total	4,250	598	14.07%

Key Findings

- Customers making four or more customer service calls have a churn rate of 50.75%.
- Customers making fewer than four calls have a churn rate of 10.93%.
- The difference is approximately 39.82 percentage points.
- The 4+ calls group represents only 7.88% of customers but accounts for approximately 28.43% of churned customers.

Business Interpretation

- Frequent customer service interactions are strongly associated with churn.
- Repeated interactions may indicate unresolved issues or customer dissatisfaction.

Recommendation

The company should investigate customers with frequent support interactions and consider proactive issue resolution, follow-up communication, and service improvement initiatives.

ANALYSIS 6: Service Calls and International Plan

Business Question

How does churn vary when customer service calls and international plan subscription are analyzed together?

Methodology

A combined PivotTable was created using Service_Call_Group and international_plan.

Results

Service Calls	International Plan	Customers	Churned	Churn Rate
0–3	No	3,553	287	8.08%
0–3	Yes	362	141	38.95%
4+	No	301	144	47.84%
4+	Yes	34	26	76.47%

Key Findings

Customers with four or more service calls and an international plan have a churn rate of 76.47%.

Customers with fewer service calls and no international plan have a churn rate of only 8.08%.

The difference between these segments is approximately 68.39 percentage points.

Business Interpretation

- Customers exhibiting both characteristics have particularly high observed churn rates.
- However, the highest-risk group contains only 34 customers, making its estimated churn rate less stable than estimates based on larger groups.

Recommendation

Prioritize further investigation of international plan subscribers who frequently contact customer support.

ANALYSIS 7: Calling Patterns vs. Churn

Business Question

Do churned customers have different calling patterns compared with retained customers?

Methodology

Average daytime, evening, nighttime, and international calling minutes were calculated for churned and retained customers.

Results

Calling Category	Retained	Churned	Difference
Daytime Minutes	175.56	208.99	+33.43
Evening Minutes	198.57	209.96	+11.39
Nighttime Minutes	199.58	206.33	+6.75
International Minutes	10.19	10.63	+0.44

Key Findings

- Churned customers have higher average calling minutes across all four categories.
- The largest difference occurs in daytime calling.
- Churned customers use approximately 19.04% more daytime minutes than retained customers.

Business Interpretation

- Higher daytime calling usage is associated with churn.
- This relationship may warrant further investigation into customer usage patterns, plan suitability, and billing expectations.

ANALYSIS 8: International Calling Usage vs. Churn

Business Question

Is international calling usage associated with customer churn?

Methodology

Average international calling minutes and charges were compared between churned and retained customers.

Results

Metric	Retained	Churned
International Minutes	10.19	10.63
International Charges	2.75	2.87

Key Findings

- Churned customers average 0.44 more international calling minutes.
- Their average international charges are approximately 0.12 higher.
- These differences are relatively small.

Business Interpretation

- International calling volume shows only a small descriptive difference between churned and retained customers.
- In contrast, international plan subscription shows a much larger difference in churn rates.
- This suggests that international plan characteristics deserve further investigation.

ANALYSIS 9: Calling Charges vs. Churn

Business Question

Do churned customers incur higher average calling charges?

Methodology

Average charges were calculated for daytime, evening, nighttime, and international calls.

Results

Charge Category	Retained	Churned	Difference
Day Charges	29.84	35.53	+5.69
Evening Charges	16.88	17.85	+0.97
Night Charges	8.98	9.29	+0.31
International Charges	2.75	2.87	+0.12

Key Findings

- Churned customers have higher average charges in all four categories.
- The largest difference occurs in daytime charges.
- Churned customers incur approximately 19.07% higher average daytime charges.

Business Interpretation

- Higher calling charges may be associated with higher usage rather than higher prices per minute.
- Further analysis would be required to determine whether plan pricing or affordability contributes to customer dissatisfaction.

Recommendation

Evaluate the suitability of existing calling plans for customers with high daytime usage.

ANALYSIS 10: Geographic Churn Analysis

Business Question

Which states have the highest and lowest observed customer churn rates?

Methodology

Customers were grouped by state.

Customer counts, churned counts, and churn rates were calculated and sorted.

Results: Highest-Churn States

State	Customers	Churned	Churn Rate
New Jersey	96	26	27.08%
California	39	10	25.64%
Washington	80	18	22.50%
Maryland	86	19	22.09%
Montana	80	17	21.25%

Results: Lowest-Churn States

State	Customers	Churned	Churn Rate
Virginia	100	5	5.00%
Hawaii	77	4	5.19%
Alaska	61	4	6.56%
Nebraska	73	5	6.85%
Rhode Island	87	6	6.90%

Key Findings

- New Jersey has the highest observed churn rate at 27.08%.
- Virginia has the lowest observed churn rate at 5.00%.
- The difference between these states is 22.08 percentage points.

Business Interpretation

Churn rates vary considerably across geographic locations.

However, relatively small state-level customer counts mean that these differences should be interpreted cautiously.

Potential areas for investigation include service quality, customer support, competition, and regional customer characteristics.

ANALYSIS 11: High-Risk Customer Segmentation

Business Question

Which customer segments have the highest churn rates when service calls, international plans, and voicemail subscriptions are analyzed together?

Methodology

Customers were segmented using three variables:

1. Service_Call_Group
2. international_plan
3. voice_mail_plan

A PivotTable was used to calculate customer counts, churned counts, and churn rates for each segment.

Results

Service Calls	International Plan	Voicemail Plan	Customers	Churned	Churn Rate
0–3	No	No	2,621	267	10.19%
0–3	No	Yes	932	20	2.15%
0–3	Yes	No	268	114	42.54%
0–3	Yes	Yes	94	27	28.72%
4+	No	No	225	116	51.56%
4+	No	Yes	76	28	36.84%
4+	Yes	No	24	19	79.17%
4+	Yes	Yes	10	7	70.00%

Key Findings

Highest Observed Churn Rate

Customers with four or more service calls, an international plan, and no voicemail plan have the highest observed churn rate of 79.17%.

However, this segment contains only 24 customers.

High-Risk Segment With Greater Churn Volume

Customers with four or more service calls, no international plan, and no voicemail plan have a churn rate of 51.56%.

This segment contains 225 customers, including 116 churned customers.

It accounts for approximately 19.40% of all churned customers.

Lowest Observed Churn Rate

Customers with fewer than four service calls, no international plan, and a voicemail plan have a churn rate of 2.15%.

Business Interpretation

The analysis demonstrates the value of combining multiple customer characteristics rather than examining each variable independently.

Both churn rate and segment size should be considered when identifying retention priorities.

ANALYSIS 12: Customer Retention Strategy

Business Question

What customer retention strategies can be recommended based on the analytical findings?

Methodology

Findings from the previous 11 analyses were consolidated to identify areas requiring business attention.

Recommended Retention Strategies

Priority	Analytical Finding	Recommended Action
High	4+ service calls: 50.75% churn	Proactive support and issue resolution
High	International plan: 42.17% churn	Review plan pricing, benefits, and complaints
Medium	Higher daytime usage among churned customers	Evaluate usage-based calling packages
Medium	No voicemail: 16.44% churn	Investigate customer engagement differences
Investigate	New Jersey: 27.08% churn	Examine regional service quality and customer experience

Strategy 1: Proactive Customer Support

Identify customers who make repeated customer service calls.

Investigate unresolved complaints and consider timely follow-up communication.

Strategy 2: International Plan Evaluation

Review international plan pricing, benefits, customer expectations, and support experiences.

Evaluate whether plan improvements could address customer concerns.

Strategy 3: Personalized Calling Plans

Investigate whether customers with high daytime usage would benefit from alternative calling packages.

Test plan recommendations before implementing them broadly.

Strategy 4: Customer Engagement

Examine whether voicemail subscription reflects broader customer engagement patterns.

Consider appropriate engagement initiatives for customers with limited service adoption.

Strategy 5: Targeted Retention Initiatives

Use combinations of customer characteristics to identify segments for further investigation.

Prioritize segments based on both observed churn rates and the number of customers affected..

6. Excel Dashboard Overview

A customer churn dashboard was developed to communicate the most important analytical findings.

6.1 Dashboard KPIs

KPI	Value
Total Customers	4,250
Churned Customers	598
Retained Customers	3,652
Overall Churn Rate	14.07%

6.2 Dashboard Visualizations

The dashboard includes five charts:

1. **Customer Churn Distribution:** Doughnut chart showing retained and churned customers.
2. **Customer Tenure vs. Churn:** Combo chart comparing tenure groups and churn rates.
3. **Churn Rate by Service Calls:** Column chart comparing customers with 0–3 and 4+ customer service calls.
4. **Churn Rate by International Plan:** Bar chart comparing international plan subscribers and non-subscribers.
5. **Churn Rate by Customer Segment:** Horizontal bar chart comparing eight customer segments.

The dashboard provides a consolidated view of customer churn patterns and supports communication of findings to business stakeholders.

---

7. Consolidated Key Insights

The analysis identified the following major insights:

1. Overall Customer Churn

598 of 4,250 customers churned, resulting in a churn rate of 14.07%.

2. Customer Service Interactions

Customers making four or more service calls exhibit substantially higher churn rates.

3. International Plan Subscription

International plan subscribers have significantly higher observed churn rates than non-subscribers.

4. Voicemail Subscription

Voicemail subscribers exhibit lower observed churn rates across the analyzed customer groups.

5. Calling Behavior

Churned customers have higher average daytime calling minutes and charges.

6. Geographic Variation

Observed churn rates differ across states, although regional sample sizes limit the reliability of individual rankings.

7. Customer Segmentation

Combining service calls, international plans, and voicemail subscriptions reveals customer groups with substantially different churn rates.

CONCLUSION

The analysis of 4,250 customer records identified an overall churn rate of 14.07%.

The strongest observed associations with churn involved frequent customer service calls and international plan subscriptions.

Additional differences were observed across voicemail subscriptions, calling behavior, charges, account tenure, and geographic locations.

Customer segmentation further demonstrated that combining multiple characteristics can reveal groups with substantially different churn rates.

Based on these findings, the project recommends investigating frequent customer service interactions, reviewing international plan offerings, evaluating calling-plan suitability, and developing targeted retention initiatives.

While the analysis does not establish the causes of customer churn, it provides a structured foundation for further investigation and evidence-based decision-making.